TWENTY

Brevity

Be brief.

Now, that was fun to write, but if advising writers to "be brief" was all it took, you and I could both just skip this chapter. We can't. I've reviewed, formally or informally, somewhere around a thousand manuscripts over my career, and all but a handful should have been shorter. This is why editors nearly always ask writers to condense their manuscripts, and why submissions, conference abstracts, and term papers have length limits. We've all been advised to be brief, but most of us still need help getting there.

There are three major reasons for this call to brevity. First, it helps publishers: journal pages are expensive to produce, and journals are deluged with submissions. Shortening each paper means journals can publish more of them. Second, brevity helps readers: all of us are swamped with written material competing for our attention (chapter 1), and shortening each paper means we can read more of what's out there. Third, brevity helps writers: shorter writing tends to be clearer writing, so your goal of effortless communication can almost always be advanced by shortening your draft.

What exactly do I mean by brevity? Manuscript lengths are most often expressed as word counts, and it's easy to overemphasize those. It's actually better to count characters than words, because replacing long words with short ones is just as helpful to the reader as reducing word count. Second, counts are only a convenient shortcut. What really matters is not how long a sentence or paragraph is, but how quickly your reader can read and understand it. If adding a little extra text actually facilitates reading, go for it; but keep in mind that removing text is nearly always a more important challenge.

Two Ways to Be Brief

There are two ways to achieve brevity in writing: reduce content, or reduce the text used to convey content. Good writers pay attention to both.

Reducing content sounds like a sacrifice, but if you think of it as sticking to your story, it's easier to appreciate its virtues. If a section, a sentence, a detail, a graphic, or a dataset isn't necessary to the story you're telling, take it out (or better still, don't put it in to begin with, which is the point of outlining and related techniques; see chapter 7). This can pay off strikingly. I recently reviewed a manuscript that identified in its Abstract a simple hypothesis and some methods to test it—but spent most of its Methods and Results, and a good chunk of its Discussion, on unnecessary details of the study system and a pair of completely unrelated experiments. Just by sticking to their story, the authors could have cut the manuscript by at least forty percent and set themselves up to have more impact on more readers. Don't miss this kind of easy opportunity.

You don't have to cut content, though, to cut length. It's amazing how much can be cut from most drafts without removing any information. If you've written a two-hundred-word paragraph, ask yourself if you can convey the same content in 190 words, or 180, or 150. Carefully consider each sentence, each phrase, and each word. If it isn't necessary, take it out. If there's a shorter alternative that's equally clear, use it. Look especially for the common text-bulgers addressed in the next section. Be ruthless: every character of your text should be forced to justify its existence.

These two approaches to brevity are best taken at different stages of the writing process. Work at sticking to your story before and as you write: there's no sense laboring to draft content that doesn't belong in your manuscript. In contrast, don't worry too much about economy of expression until you've completed a first draft, lest you lose writing momentum (chapter 6). You can shorten during self-revision (chapter 21).

Common Text-Bulgers

There isn't really any limit to the list of ways that a manuscript can become overlong. However, it's possible to identify some common text-

bulging habits. Put more positively, these are common and easy opportunities to improve brevity.

- **The passive voice**. The active voice is shorter, along with its other advantages (chapter 18).
- **Nominalizations**. Turning verbs into nouns makes writing both weaker and longer (chapter 19). You can *reduce* length by not writing "*achieve a reduction in* length."
- **Long words**. Whether technical or commonplace, a long word belongs only when no shorter one will do (chapter 19). Don't *endeavor* to do something you can *try*, or *terminate* something you can *end*.
- **Roundabout phrases**. Many common phrases can be replaced by single words that cut more quickly to the point: for example, *owing to the fact that* (=*because*), *notwithstanding the fact that* (=*although*), *is able to* (=*can*), *the majority of* (=*most*).
- **Tautologous modifiers**. Watch for modifiers already implied by the words they modify: *completely finish*, *may potentially*, *ultimate result*, *blue in color*. (Are you worried your reader will think the crystals were blue in shape?)
- **Empty modifiers**. Some common modifiers often lack meaning: *really*, *basically*, *actually*, *indeed*, *quite*, *various*. While these words can sometimes help a sentence, they're often just verbal tics fossilized in ink.
- **Padding**. Sometimes an entire phrase bulks up a sentence without adding information: *the fact of the matter is*, *in our opinion*, *needless to say*, *it obviously follows that*, *for all intents and purposes*. In speech such phrases allow time for the listener to catch up, or for the speaker to think of what to say next. In writing, neither function applies. Delete!
- **Hedging**. It's essential to use hedges (chapter 13) to express the precise strength (or limitations) of the claims you're making. Unfortunately, hedges are like potato chips: once you write one, it's hard to stop, and they expand your text like your waistline. As an example, I read this in a thesis once: "(Observation) could reasonably be assumed to possibly occur by chance." One level of hedging would have done—perhaps "The evidence that (observation) was nonrandom was modest ($P = 0.06$)." Excessive hedging adds length, and also makes it difficult for readers to take your argument seriously. Evaluate each hedge, and keep only those necessary to avoid misleading readers about the strength of your conclusions. Similar logic applies to the use of emphatics, such as *clearly*, *primary*, and *major*.

- **Metadiscourse**. This is writing that's about the writing rather than about the subject of the manuscript (Williams 1990:40). A piece of metadiscourse may refer to the structure and content of the text ("In this section we report . . ."), to the writer's thoughts about or rationale for the writing ("We believe that . . .", or "It has been observed that . . ."), or even to the reader's act of reading ("Consider the following . . ."). Like hedging, metadiscourse can be used effectively: for instance, to provide organizational cues for the reader. But it can also proliferate. Check metadiscourse skeptically, and remove any that only adds length. For instance, "We believe that our results establish that salinity is . . ." says no more than "Salinity is . . ." Other common excesses of metadiscourse include "In this study we" (just say "We . . ."), "The objective of this study was to . . ." ("We sought to . . .", or just "We . . ."), and "It is important to keep in mind that . . ." (omit).
- **Parentheticals**. A parenthetical is a phrase that interrupts an otherwise complete sentence (or a sentence that interrupts an otherwise complete paragraph). As the name suggests, many are enclosed by parentheses, but they can also be set off by commas, by dashes, or not at all. For example, this passage includes three parentheticals:

  "Many recovered meteorites are of the nickel-iron type *in part because these are easily recognized as geologically unusual*. However, most meteorites are actually chondrites. Carbonaceous chondrites, *as opposed to the more common ordinary chondrites*, often contain complex organic molecules *such as amino acids*."

  Parentheticals can clarify, illustrate, restrict, or otherwise modify meaning. They can also offer extra information that makes the material more interesting or more useful, even if it isn't crucial. However, parentheticals add length, interrupt the reader's flow through the logic of your argument, and distract the reader from the story you're telling. Identify each parenthetical and ask whether it's really worth the demand it places on your reader[1].

[1] This book includes quite a few parentheticals (especially footnotes like this one, and how's that for metadiscourse?). In a shorter writing form I'd have excised them ruthlessly. In a book, where

- **Redundancy.** It's amazing how easily redundancy creeps into writing. Check for redundant content throughout your manuscript, but pay special attention to three common trouble spots:
  - Keeping paragraphs to the intended outline. Each paragraph has its own topic, and should stick to it. Once you've covered a topic, don't go back to it in later paragraphs. If you find that you need to, it probably signals a need for revision to your outline.
  - Separating the content of major sections. It's tempting to repeat details of the Methods in the Results section, and to repeat Results in the Discussion. A few cross-references are indeed useful to readers (chapters 12–13), but be sparing. The Introduction and Discussion may have more overlap, because the Introduction sets up questions to be answered in the Discussion, and because both sections need to address things like the context, goals, and significance of the study. The trick is for the Discussion to *refer to* material from the Introduction while *repeating* as little of it as possible.
  - Efficient use of text, tables, and figures. Remember that there's a best way to present any particular dataset or pattern (chapter 12)—but one way will suffice. Don't present the same data in a table and in a figure, for instance. Text referring to a figure or table should provide just enough description that the reader knows what pattern (s)he is being asked to see: for instance, "reaction yield was inversely related to temperature (Figure 1)." More detailed description is wasted.

Getting to Brevity: An Example

To illustrate some of these text-bulgers in a real piece of scientific writing, here's a conference Abstract written by my graduate student Chandra Moffat. It concerns the suitability of a European insect (gall wasp) to control a pest plant (hawkweed) that's been introduced to North America. The central question is whether any collection of the wasp would do, or whether wasps collected from different places or different plant hosts vary in ways that would affect their usefulness for biological control. A revision deliberately targeting length took it from the "draft" version at 394 words to the "short" version at 276 words: thirty percent shorter, and noticeably easier to read (even if you don't understand the technical content). Here they are:

writing need not be quite so lean, there is room for debate about footnotes and other parentheticals. Sword (2012) deplores the frequent use of footnotes as "the spilled sewage of excessive marginalia," but I chose to use them to share some reasons that thinking about writing needn't be dull. Some readers have loved my footnotes, and some have hated them. I apologize if you've hated them.

Is cryptic diversity in the *Pilosella* leaf-gall wasp associated with geography, host plant, or *Wolbachia* infection?

Draft version:

Biological control programs require accurate assessments of the host-range of agents prior to introduction. Discoveries of cryptic genetic variation of specialist herbivorous insects are being made at a dramatic rate, and have appeared in a few weed biological control systems, but only post-introduction. Here, we present the first study, to our knowledge, showing cryptic genetic differentiation in a candidate weed biocontrol agent prior to release.

The gall wasp *Aulacidea pilosellae* is a candidate biocontrol agent for multiple species of European hawkweeds (*Pilosella*) that are invasive in North America. Preliminary surveys in the wasp's native range in Europe suggested that it has both Northern and Southern biotypes, which appear to differ in host range, voltinism, and reproductive mode. We performed thorough and widespread surveys on multiple host species in four distinct geographic areas and sequenced three gene regions (CO1, 28S, ITS-II) to determine (i) whether there was any genetic evidence supporting the hypothesis of multiple biotypes, (ii) whether variation was based on geographic separation (as we predicted) or on host-plant association, and (iii) whether any individuals were infected with the bacterial endosymbiont *Wolbachia*, which is known to alter a number of life history characteristics in Hymenoptera.

We found considerable genetic divergence among populations of *A. pilosellae*, providing genetic support for the hypothesis that this species has multiple biotypes. This variation was found in the CO1 region, but not in the nuclear regions sequenced. A MrBAYES 50% majority-rule consensus phylogeny organized the populations sampled into three dis-

tinct lineages. While we predicted that any variation would be based on geographic separation of populations, the lineages clustered our populations into host-associated groups. Individuals collected from the host *P. officinarum* in the Northern Range clustered with individuals collected from the same host in the Southern Range, rather than with individuals from nearby collection sites which were found on different host species. Finally, only a few populations tested positive for infection with the endosymbiont *Wolbachia*: all of these were sampled from populations clustered in a single lineage. Our results have major implications for the biocontrol of *Pilosella* hawkweeds. Prior to our study, differences in biology between biotypes were attributed to geography. In contrast, we found that biotypes correspond to host association. These results demonstrate the value of genetic typing among conspecific populations of biocontrol agents to better define host-associations and reduce the risk of non-target attack prior to agent release.

Short version:

Biological control programs require accurate assessments of the host-range of agents before introduction. It is increasingly clear that cryptic, host-associated genetic variation is common in specialist herbivorous insects, but for weed biocontrol agents genetic structure has been reported only for a few post-introduction systems. We demonstrate, for the first time, cryptic genetic differentiation in a candidate agent prior to its release.

The gall wasp *Aulacidea pilosellae* is a candidate agent for biocontrol of several European hawkweeds (*Pilosella*) that are invasive in North America. Preliminary surveys in the wasp's native range in Europe suggested that it has Northern and Southern biotypes differing in voltinism, host range, and reproductive mode. We surveyed wasps on multiple host species in four geographic areas and sequenced three gene regions (CO1, 28S, ITSII) to determine (i) whether there was genetic evidence for multiple biotypes, and (ii) whether genetic variation was associated with geographic separation, host-plant association, and/or infection with the bacterial endosymbiont *Wolbachia* (known to alter reproductive mode in Hymenoptera).

We found considerable genetic divergence among *A. pilosellae* populations, supporting the hypothesis of multiple biotypes, although this reflected variation only in CO1 and not the nuclear regions. A 50% majority-rule consensus phylogeny suggested three distinct lineages, which primarily corresponded with host association, not geography. *Wolbachia* infection was found in several populations, all of which were grouped in one host-associated lineage. Our results have important implications for hawkweed biocontrol, because they suggest that different wasp accessions will better target different members of the invasive *Pilosella* complex. Our results demonstrate the value of genetic typing in source populations of biocontrol agents, in order to define host associations and reduce the risk of non-target attack.

Note that you might not have produced the same "short" version as Chandra did: there are always choices, and your version might have stressed a slightly different story or told that story a bit differently.

It's important to point out that I didn't choose this example because Chandra is a terrible writer producing unusually bloated drafts. In fact, this is a typical example of a good writer working at her craft. As most writers should, she chose momentum over polish in the first draft, then improved what she'd written during self-revision (see chapter 21) and by taking advantage of friendly review (see chapter 22).

A final lesson here is that each writer has their own habits. Chandra's draft Abstract, for example, was relatively rich in redundancy and roundabout phrases, but largely avoided excess hedging. Your own writing may be different: perhaps you have a fondness for parentheticals, or metadiscourse, or padding phrases. Don't be ashamed of these habits—we all have them. Instead, think of familiarity with your habits as an easy way to find opportunities to streamline your text.

Chapter Summary

- Brevity in writing benefits readers, publishers, and writers.
- Most writers have habits that work against brevity—for instance, use of the passive voice, roundabout phrases, empty modifiers, and excessive hedging and metadiscourse.
- Because momentum is so important in writing the first draft, achieving brevity is a matter for revision. Dramatic reductions in length should be routine.